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Educational Course: Clustering and the k-means Algorithm

Understanding the Clustering Problem

The "Natural Grouping" Metaphor

Imagine you're looking at the starry sky. Some stars seem naturally grouped into **constellations**. No one told you which stars form the Big Dipper - you see it for yourself.

This is exactly what **clustering** is: discovering natural groups in data, without prior guidance.

The Essence of k-means

The Dual Mission

k-means seeks to accomplish **two simultaneous objectives**:

1. **Find K representative centers**
2. **Assign each point** to the nearest center

But how do we judge if we've succeeded?

The Objective Function: The Mathematical Heart

The Intuitive Idea

Let's measure how far each point is **from the center** of its group. The smaller this distance, the better the grouping.

For the entire dataset, we calculate:

$$\text{Total error} = \sum_{\text{all points}} (\text{distance to its group center})^2$$

Mathematical Formalization

Let's define precisely:

- **Data**: N points $\mathbf{x}_1, \mathbf{x}_2, ..., \mathbf{x}_N$ in a D-dimensional space
- **Clusters**: K groups (number chosen in advance)
- **Centers**: $\mu_1, \mu_2, ..., \mu_K$ (to be determined)
- **Assignment**: For each point i and cluster k, $z_{ik} = 1$ if point i is in cluster k, 0 otherwise

The Function to Minimize

The objective function (or cost function) is:

$$J = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \|\mathbf{x}_i - \mu_k\|^2$$

Let's break down this formula:

Symbol	Meaning	Role in the formula
z_{ik}	Assignment indicator	Selects only the cluster to which the point belongs
$\|\mathbf{x}_i - \mu_k\|^2$	Squared distance	Measures how far the point is from the center
$\sum_{i=1}^{N}$	Sum over all points	Accumulates error for the entire dataset
$\sum_{k=1}^{K}$	Sum over all clusters	Considers all possible clusters

In practice: Since each point belongs to only one cluster, for a point $\mathbf{x}_i$, only one z_{ik} equals 1, the others equal 0. So for each point, we only count the distance to its own center.

Why Minimize THIS Specific Function?

Reason 1: Geometric Interpretation

If we fix the assignments z_{ik}, then for each cluster k:

$$J_k = \sum_{i: z_{ik}=1} \|\mathbf{x}_i - \mu_k\|^2$$

This is exactly the **variance** of the points in cluster k, multiplied by the number of points.

Minimizing J = Minimizing intra-cluster variance = Making clusters more compact

Reason 2: An Analytical Solution Exists

If we differentiate J with respect to μ_k:

$$\frac{\partial J}{\partial \mu_k} = -2 \sum_{i=1}^{N} z_{ik}(\mathbf{x}_i - \mu_k)$$

Setting the derivative equal to zero:

$$\sum_{i=1}^{N} z_{ik}(\mathbf{x}_i - \mu_k) = 0$$

$$\Rightarrow \mu_k = \frac{\sum_{i=1}^{N} z_{ik}\mathbf{x}_i}{\sum_{i=1}^{N} z_{ik}}$$

Interpretation: The optimal center is the **mean** (center of gravity) of the points in the cluster.

Reason 3: Connection with Statistics

This function corresponds to the **log-likelihood** of a model where each cluster is a spherical Gaussian with the same variance.

The Optimization: How to Minimize This Function?

The Challenge

We have two types of variables:

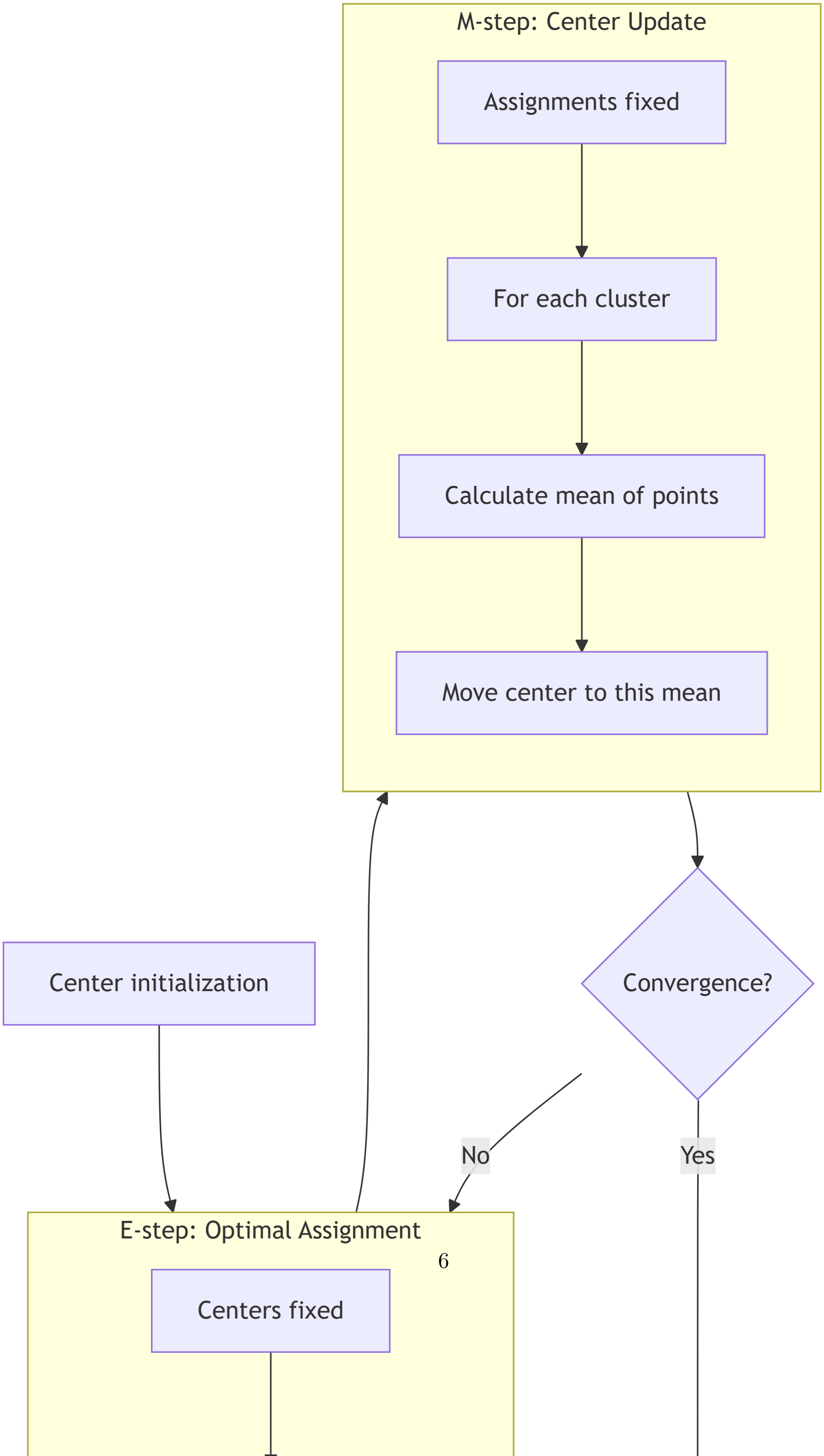
1. The **assignments** z_{ik} (discrete, binary variables)
2. The **centers** μ_k (continuous variables)

The combinatorial problem is **NP-hard**: impossible to test all combinations.

The Strategy: Coordinate Descent

Instead of optimizing all variables simultaneously, we alternate:

1. **Optimize assignments** while fixing centers
2. **Optimize centers** while fixing assignments



M-step: Center Update
Assignments fixed
For each cluster
Calculate mean of points
Move center to this mean
Center initialization
Convergence?
No
Yes
E-step: Optimal Assignment
Centers fixed

Details of the Two Optimization Steps

Step 1: Assignment Optimization (E)

Assumption: The centers $\mu_1, ..., \mu_K$ are fixed.

Objective: Choose the z_{ik} to minimize J.

Observation: The function J is a sum of positive terms. To minimize a sum, we minimize each term individually.

For a given point $\mathbf{x}_i$, we must choose a single k such that $z_{ik} = 1$. The corresponding term is $\|\mathbf{x}_i - \mu_k\|^2$.

Optimal solution: Assign $\mathbf{x}_i$ to the nearest center!

Formally:

$$z_{ik} = \begin{cases} 1 & \text{if } k = \arg\min_j \|\mathbf{x}_i - \mu_j\|^2 \\ 0 & \text{otherwise} \end{cases}$$

Complexity: For each point (N), compare with each center (K) $\rightarrow O(N \times K \times D)$ operations.

Step 2: Center Optimization (M)

Assumption: The assignments z_{ik} are fixed.

Objective: Choose the μ_k to minimize J.

Calculation: As seen previously, by differentiating with respect to μ_k:

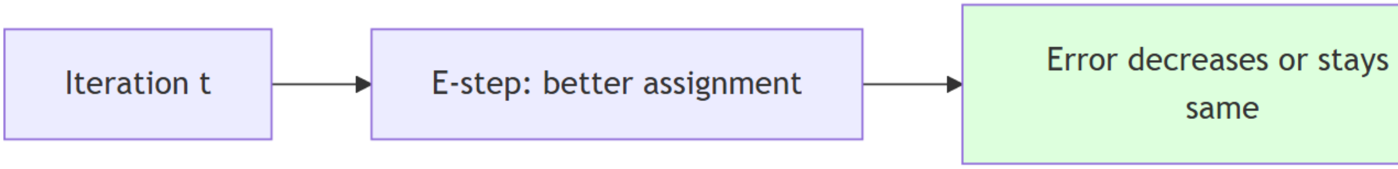
$$\mu_k = \frac{\sum_{i=1}^{N} z_{ik}\mathbf{x}_i}{\sum_{i=1}^{N} z_{ik}}$$

Interpretation: The new center of cluster k is the **mean** (centroid) of all points assigned to it.

Complexity: For each cluster (K), average its points $\rightarrow O(N \times D)$ operations.

Proof That This Alternation Always Decreases Error

A Visual Argument



Mathematical Proof

1. **E-step**: For fixed centers $\mu_k^{(t)}$, choosing $z_{ik}^{(t+1)}$ as the optimal assignment guarantees:

$$J(\mu^{(t)}, z^{(t+1)}) \leq J(\mu^{(t)}, z^{(t)})$$

   (strictly less if an assignment changes)

2. **M-step**: For fixed assignments $z_{ik}^{(t+1)}$, choosing $\mu_k^{(t+1)}$ as the optimal means guarantees:

$$J(\mu^{(t+1)}, z^{(t+1)}) \leq J(\mu^{(t)}, z^{(t+1)})$$

   (strictly less if a center changes)

3. **Conclusion**: At each complete iteration (E then M), the error can only decrease:

$$J^{(t+1)} \leq J^{(t)}$$

Important Consequence

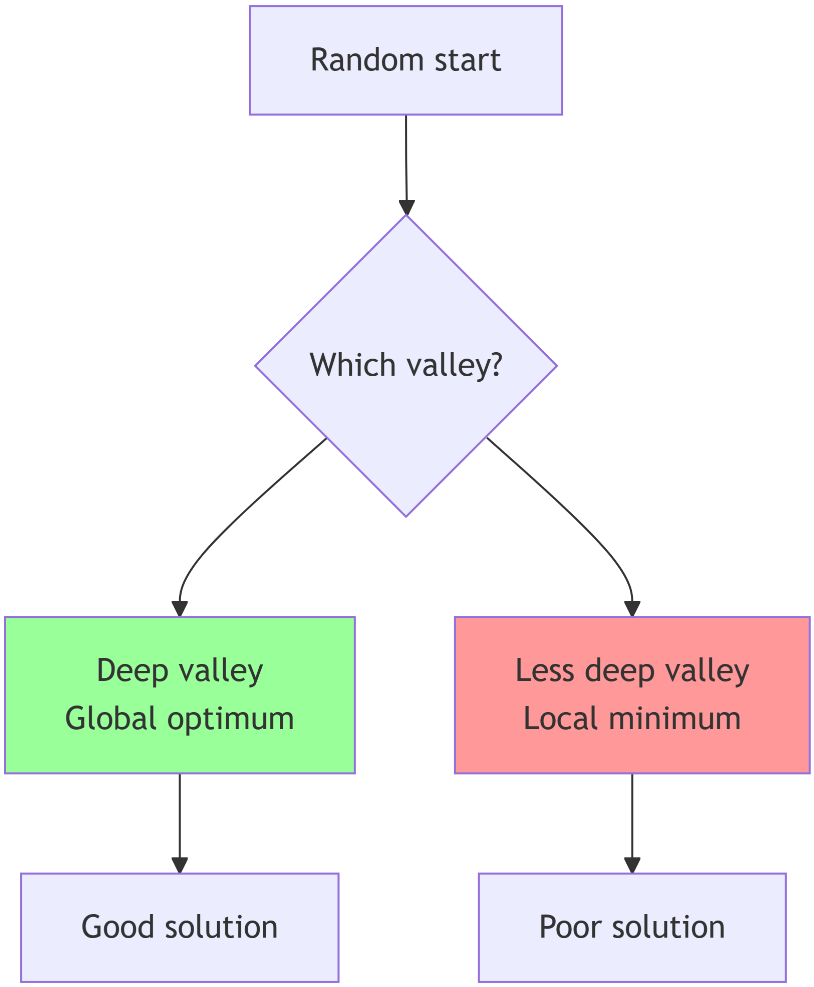
Since the error is always positive and decreases with each iteration, the algorithm **necessarily converges**.

The Trap of Local Minima

The Mountain Analogy

Imagine you're looking for the lowest point in a mountainous region, blindfolded. You always go downhill. You'll end up in a valley, but not necessarily in the lowest valley!

This is exactly what happens to k-means:



Why Does This Happen?

The function J is not **convex** with respect to all variables. It has several "valleys" (local minima).

Concrete example: With 4 points forming a square, and K=2, there are two equivalent solutions:

- Solution 1: Cluster {(1,2)}, Cluster {(3,4)}
- Solution 2: Cluster {(1,3)}, Cluster {(2,4)}

Both give the same value of J, but they are different local minima.

Smart Initialization: k-means++

The Principle

Rather than choosing initial centers completely randomly, choose them in a way that "covers" the data space.

Detailed Algorithm

1. **First center**: Chosen uniformly at random from all points
2. **For each subsequent center**:

   a. For each point $\mathbf{x}_i$, calculate:

   $$D(\mathbf{x}_i) = \min_{\text{centers already chosen}} \|\mathbf{x}_i - \mu_j\|^2$$

   (Squared distance to the nearest center already chosen)

   b. Choose the next center with probability proportional to $D(\mathbf{x}_i)^2$

   $$P(\text{choose } \mathbf{x}_i) = \frac{D(\mathbf{x}_i)^2}{\sum_{j=1}^{N} D(\mathbf{x}_j)^2}$$

3. **Repeat** until K centers

Why It Works

- Points far from existing centers have a higher probability of being chosen
- This guarantees that initial centers are **spread out** throughout the space
- In practice, this gives better results than purely random initialization

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Relation to Gaussian Models

The Implicit Assumption

k-means assumes that each cluster approximately follows a **normal** (Gaussian) distribution with:

1. **Spherical shape**: The covariance is $\sigma^2 I$ (same in all directions)
2. **Same variance**: All clusters have the same "spread"
3. **Different centers**: Each cluster has its own center μ_k

The Corresponding Probabilistic Model

If each cluster k is modeled by a Gaussian $\mathcal{N}(\mu_k, \sigma^2 I)$, then the probability density is:

$$p(\mathbf{x}|\text{cluster } k) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|\mathbf{x} - \mu_k\|^2}{2\sigma^2}\right)$$

The Mathematical Connection

Consider the **log-likelihood** of the data:

$$\log p(\mathcal{X}|\{\mu_k\}) = \sum_{i=1}^{N} \log\left(\sum_{k=1}^{K} \pi_k p(\mathbf{x}_i|\mu_k, \sigma^2 I)\right)$$

With $\pi_k = 1/K$ (equiprobable clusters).

Limiting case: When $\sigma^2 \to 0$ (very "peaked" distributions), the dominant term in the sum is that of the nearest cluster.

We then obtain approximately:

$$\log p(\mathcal{X}|\{\mu_k\}) \approx \text{constant} - \frac{1}{2\sigma^2} \sum_{i=1}^{N} \min_k \|\mathbf{x}_i - \mu_k\|^2$$

Maximizing this likelihood is equivalent to **minimizing**:

$$\sum_{i=1}^{N} \min_k \|\mathbf{x}_i - \mu_k\|^2$$

Which is exactly the function J of k-means!

Conclusion of the Connection

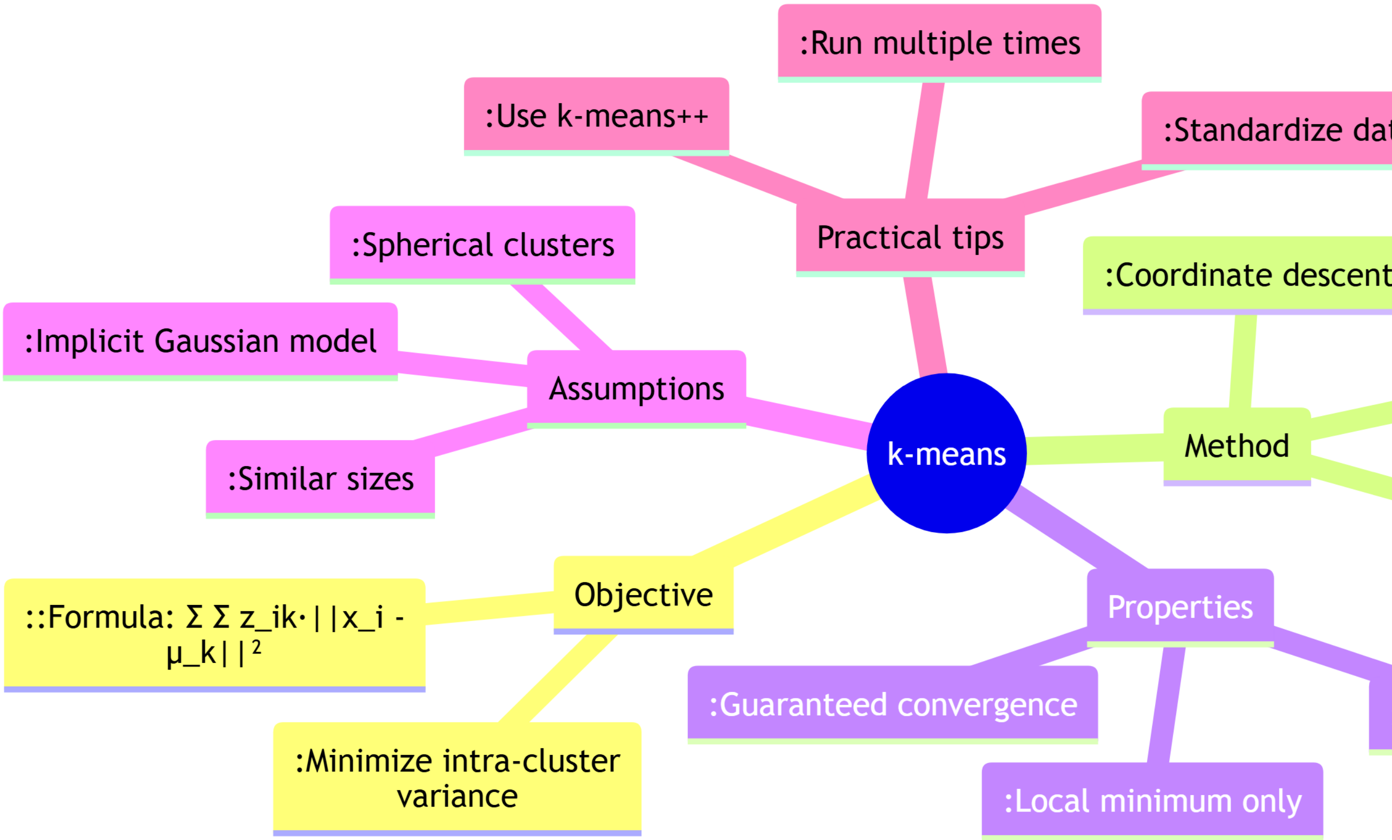
k-means can be viewed as an **EM algorithm** for a mixture of spherical Gaussians with the same variance, taking the limit where the distributions are infinitely "peaked" (hard assignment).

Pedagogical Summary

What to Remember

1. **k-means minimizes**: The sum of squared distances between each point and its cluster center
2. **Optimization method**: Alternation between two simple steps:
   - **Assignment**: Each point joins the nearest center
   - **Update**: Each center becomes the mean of its points
3. **Guaranteed property**: Error decreases at each iteration until convergence
4. **Main limitation**: Converges to a local minimum, not necessarily global
5. **Key improvement**: k-means++ for better initialization

Visual Summary



Going Further

Questions for Deepening Understanding

1. **What happens if we change the metric?** (From Euclidean distance to Manhattan distance)
2. **How to adapt k-means** for clusters with different densities?
3. **Why not minimize the sum of distances** (not squared)?

Suggested Interactive Exploration

Mentally try this exercise:

- Take 6 points forming two equilateral triangles
- Apply k-means with K=2
- Observe how initialization influences the final result

Final Word

k-means is like a **conductor** organizing musicians (the points) into sections (clusters) by placing each musician near the section leader (center) while keeping sections apart from each other.

The beauty of the algorithm lies in its **simplicity**: a simple idea (minimize distances), an effective method (alternation), and interpretable results.

Remember: Understanding the objective function and its optimization is the key to using k-means wisely and correctly interpreting its results!